

Metacommunities II

Today's Agenda:

- Quiz
- Neutral theory

Patchy Environments

Patches: suitable habitat, surrounded by unsuitable habitat.

Metapopulation: collection of connected populations

- Multiple patches connected by dispersal

Metacommunity: collection of connected communities

- Multiple patches connected by dispersal of one or more species

Metacommunities

4 Main perspectives:

- 1) Patch dynamics - extension of metapopulation stuff to more species
- 2) Species sorting - community assembly-ish, species differ in abiotic niches
- 3) Mass effects - species sorting-ish, w/ source-sink and rescue effects
- 4) Neutral theory - assumes species are identical, environment irrelevant

Metacommunities

4 Main perspectives:

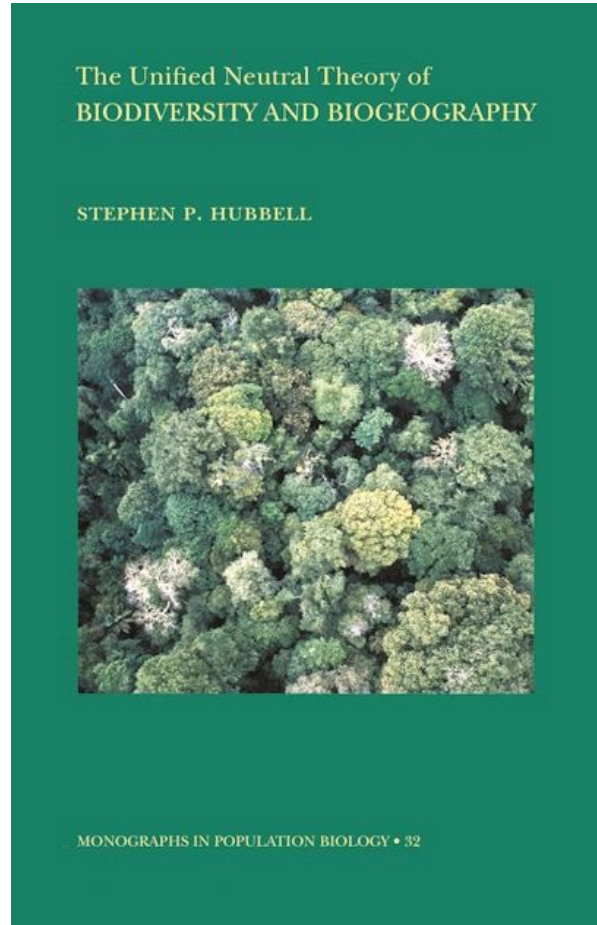
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Neutral Theory

Published 2001

Hubbell

- Tropical ecologist
- BCI in Panama
- Forest plot mapping

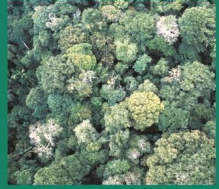




Neutral Theory

Key assumptions:

- Individuals of different species are equivalent in terms of:
 - Birth rate
 - Death rate
 - Dispersal rate
- These factors don't vary with environment



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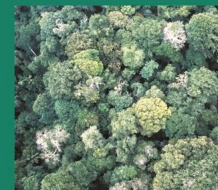
What do you think of these assumptions?
When might they be true?



Neutral Theory: Initial formulation

Initial assumptions:

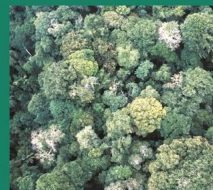
- Number of individuals constant
- Space is limiting
- All space is occupied
- Species are equally likely to colonize (proportional to abundance)
- Death occurs at a constant rate



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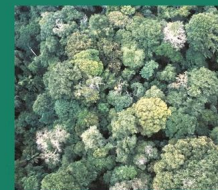
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Neutral Theory: Later formulation

Book changes:

- Incorporated metacommunities and dispersal
 - Patches can be re-colonized, slowing extinction





Neutral Theory: Later formulation

Book changes:

- Incorporated metacommunities and dispersal
 - Patches can be re-colonized, slowing extinction
- Incorporated speciation
 - Species emerge at a fixed, per-capita rate.
 - Can counteract extinctions

Testing Neutral Theory

Neutral theory makes some very testable predictions

- Different aspects of ecology
 - Species abundance distributions (and how they change)
 - Species-area relationships
 - Coexistence
 - Community assembly

Testing Neutral Theory

Neutral theory predicts a certain form of SAD

- Involves fitting a model with 3 parameters
- However, other models exist for other theories (METE, MST, etc).

McGill compared the performance of neutral model with null models

- Null models fit better
- Null models used fewer parameters (more parsimonious)

A test of the unified neutral theory of biodiversity

Brian J. McGill

Department of Ecology and Evolutionary Biology, University of Arizona, Tucson, Arizona 85721, USA

The commonness of rarity: Global and future distribution of rarity across land plants

Brian J. Enquist^{1,2*}, Xiao Feng³, Brad Boyle⁴, Brian Maltner¹, Erica A. Newman^{1,3}, Peter Møller Jørgensen⁵, Patrick R. Roehrdanz⁶, Barbara M. Thiers⁵, Joseph R. Burger⁷, Richard T. Corlett⁷, Thomas L. P. Couvreur⁸, Gilles Dauby⁹, John C. Donoghue¹⁰, Wendy Foden¹¹, Jon C. Lovett^{12,13}, Pablo A. Marquet^{14,15}, Cory Merow¹⁶, Guy Midgley¹⁷, Naia Morueta-Holme¹⁸, Danilo M. Neves¹⁹, Ary T. Oliveira-Filho¹⁹, Nathan J. B. Kraft²⁰, Daniel S. Park²¹, Robert K. Peet²², Michiel Pilonet¹, Josep M. Serra-Diaz²³, Brody Sandel²⁴, Mark Schildhauer²⁵, Irena Simová^{26,27}, Cyrille Violle²⁸, Jan J. Wieringa²⁹, Susan K. Wiser³⁰, Lee Hannah³, Jens-Christian Svenning³¹, Brian J. McGill³²

Testing Neutral Theory

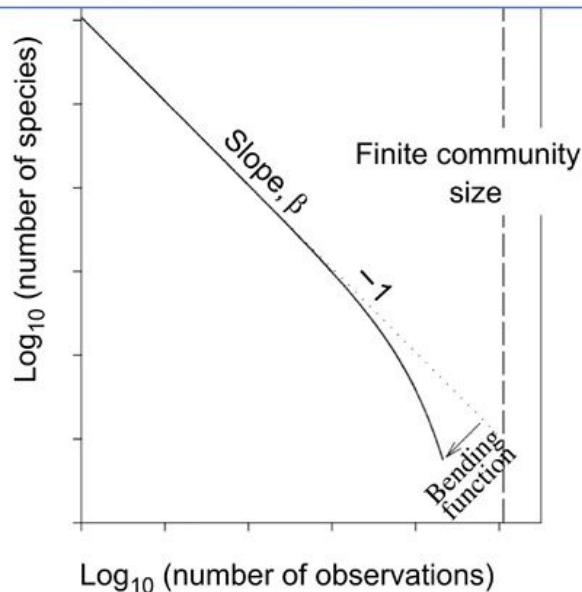
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gSAD distribution predictions:

- (i) Neutral theory and niche theory: gSAD will be best fit by log-series distribution;
- (ii) Central limit theorem: gSAD will be best fit by lognormal distribution

gSAD slope predictions:

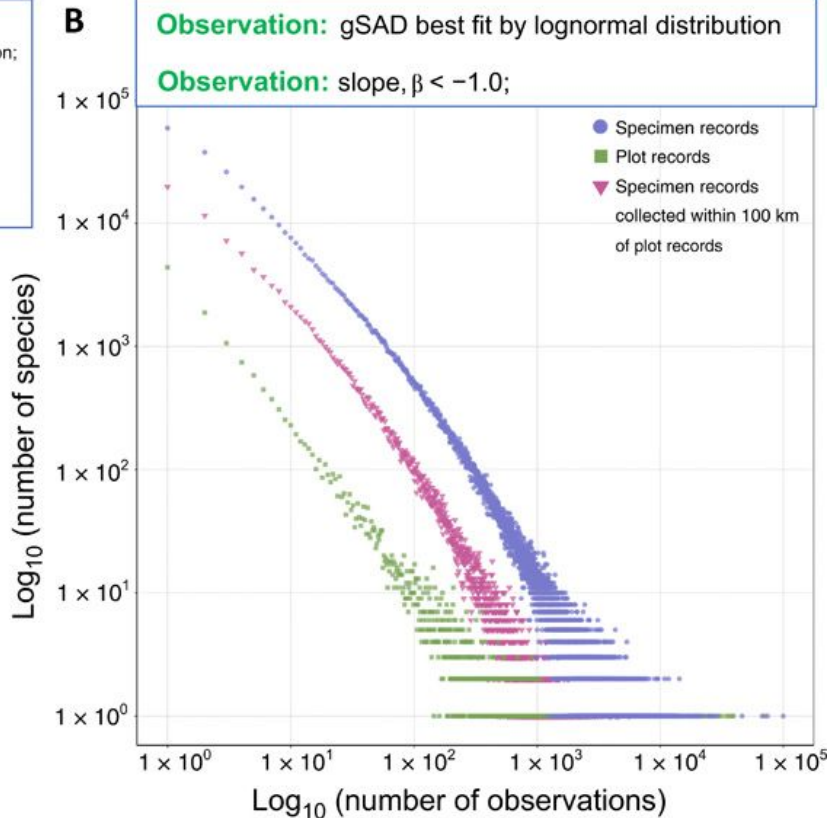
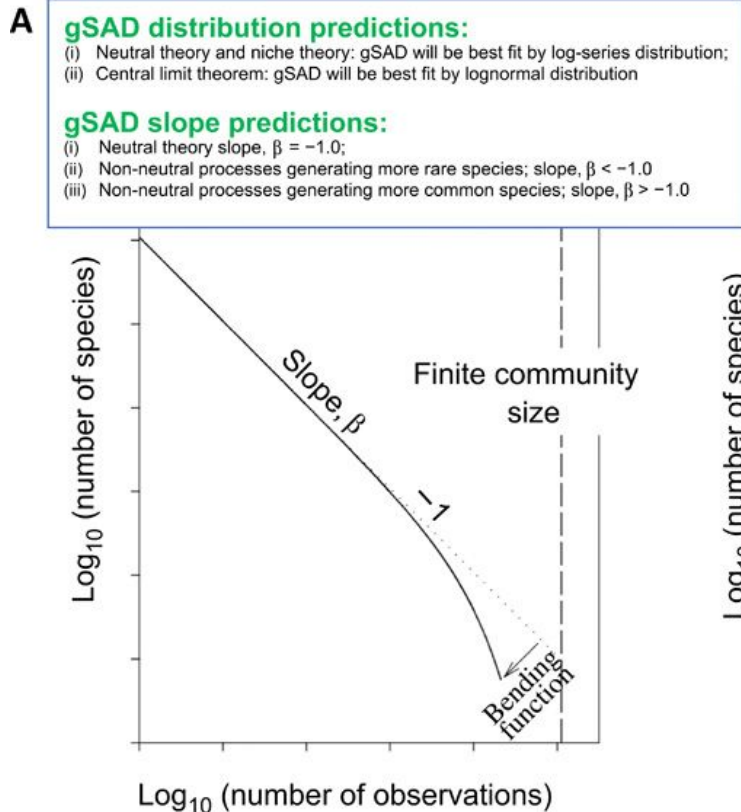
- (i) Neutral theory slope, $\beta = -1.0$;
- (ii) Non-neutral processes generating more rare species; slope, $\beta < -1.0$
- (iii) Non-neutral processes generating more common species; slope, $\beta > -1.0$



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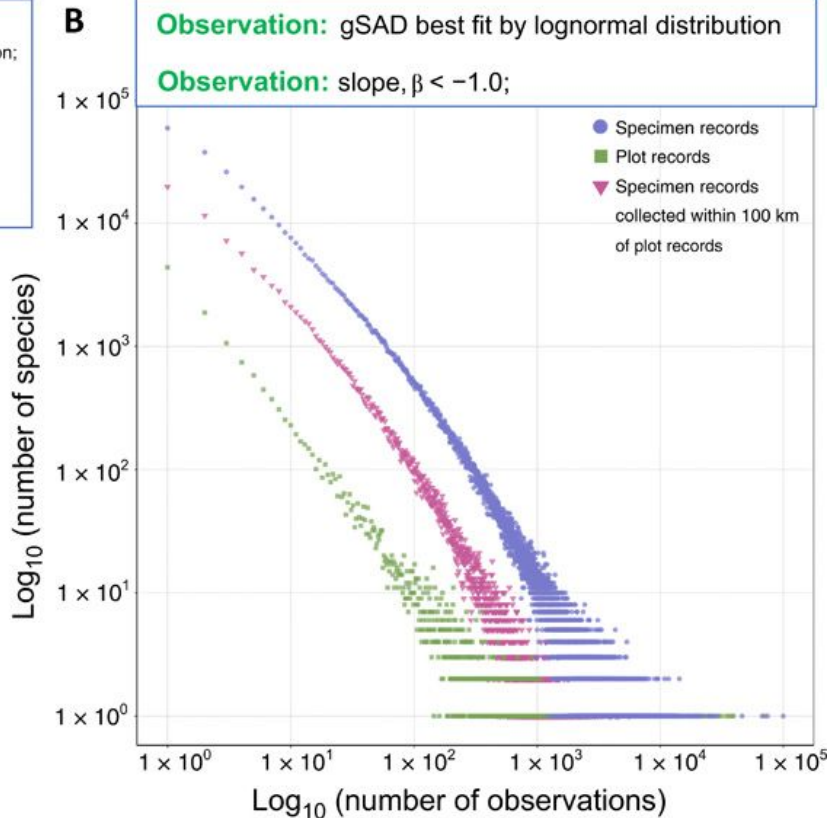
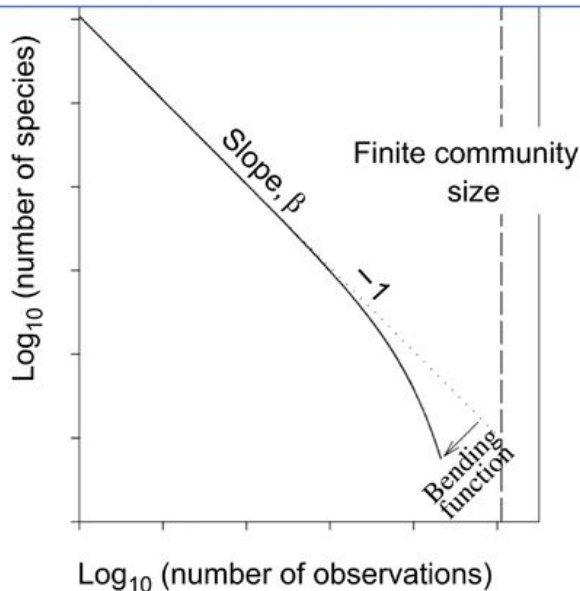


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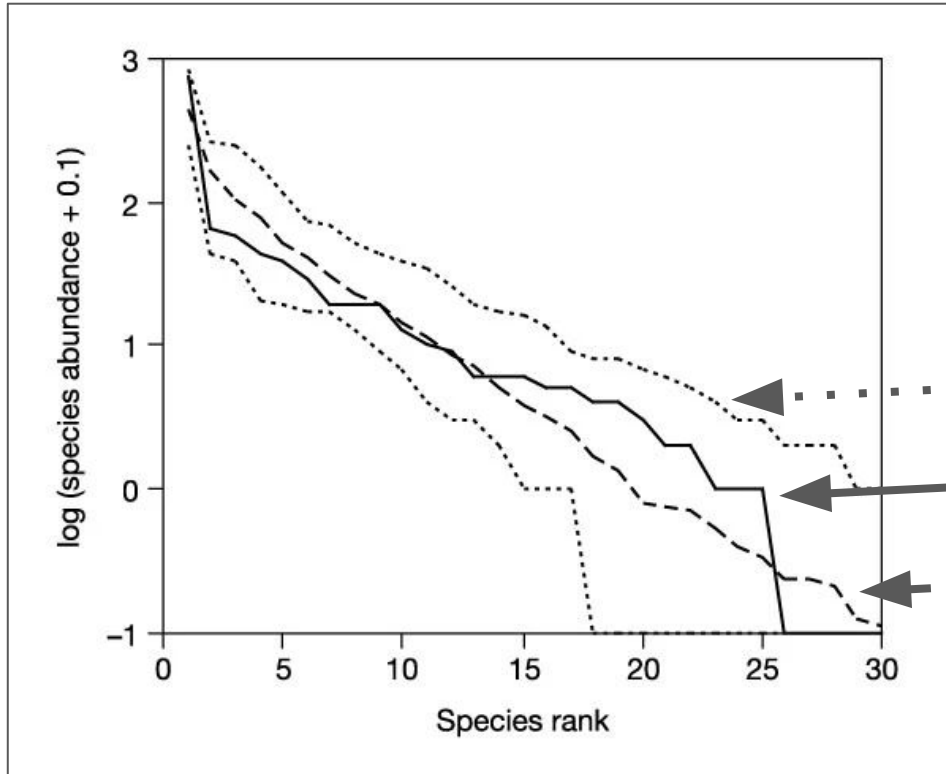
Testing Neutral Theory

Test using intertidal organisms

- 12 year record
- Lots of 60 x 60 cm quadrats

Parameterized neutral model using field data

Testing Neutral Theory



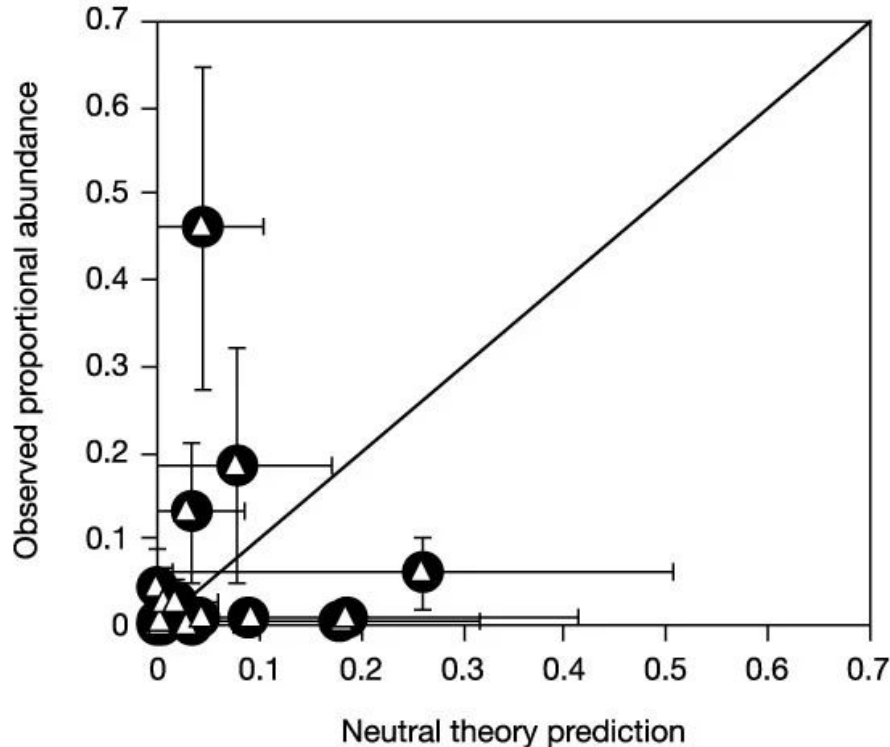
Observed SAD matched prediction

95% CI

Observed

Predicted

Testing Neutral Theory



However, deviations observed when predictions looked at dynamics over time

- So even when neutral theory gets the distribution correct, it can still get other things quite wrong

Neutral theory and coexistence

What needs to occur for species to coexist?

Can this happen under neutral theory?

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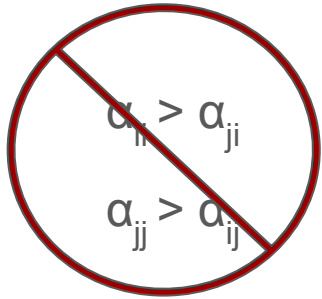
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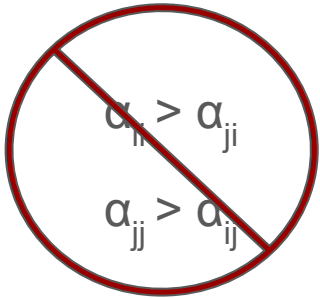


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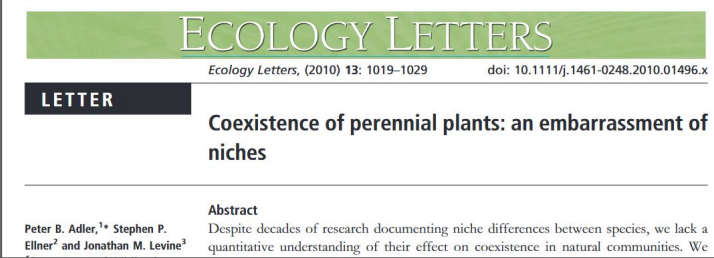
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But, under coexistence theory, even small differences can lead to competitive exclusion!

Neutral theory and coexistence

Data set of experiment focusing on sheep grazing

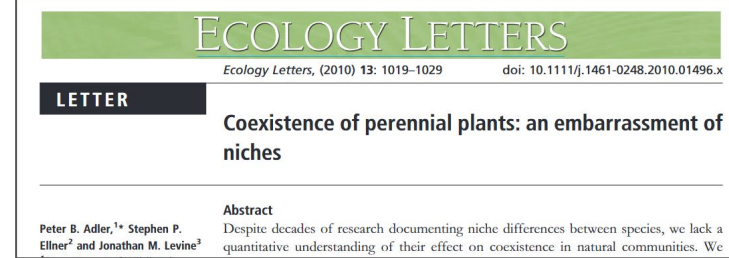
- Grazing vs no grazing
- 22 years of data
- Spatial data of individual plants recorded



Neutral theory and coexistence

Used data to parameterize a model of competition

- Model parameters can be altered to make species neutral
 - Effects on heterospecifics set to equal effect on conspecifics
 - Allows “experimenting” on things you couldn’t actually manipulate!

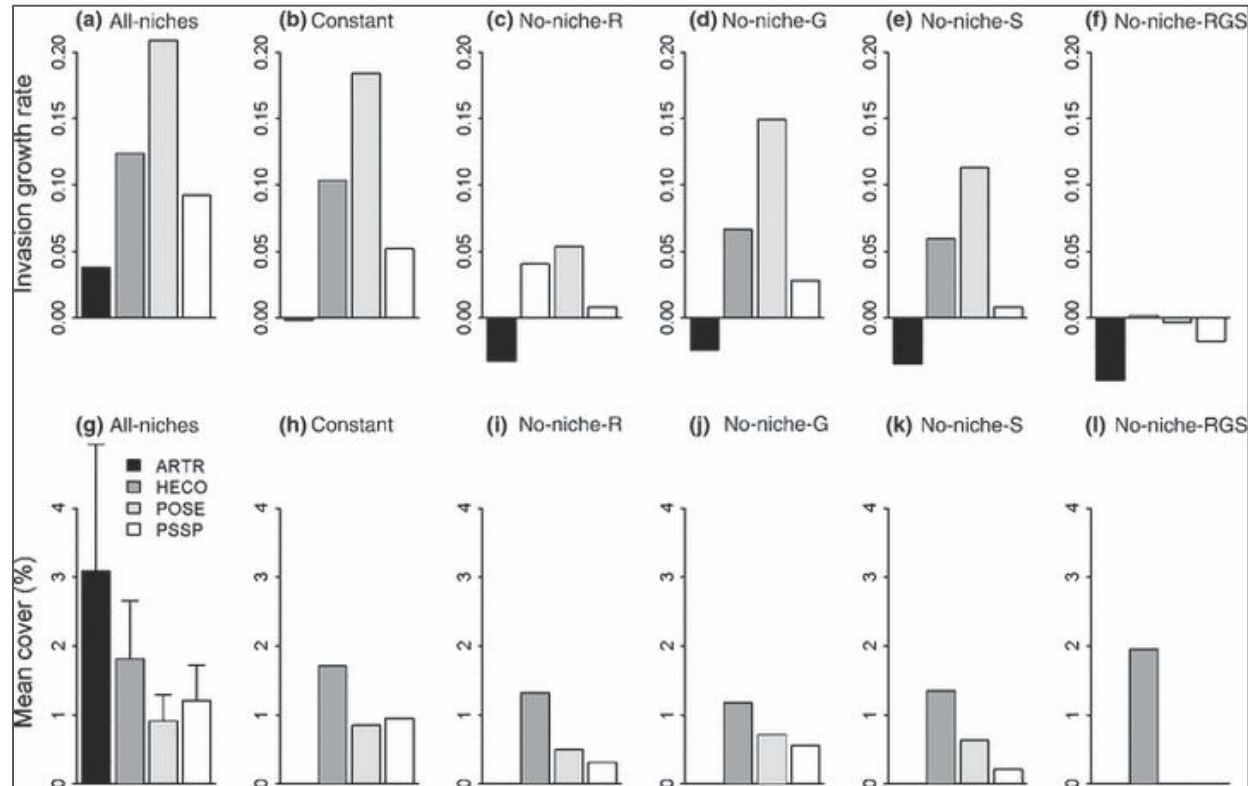


Peter B. Adler,^{1*} Stephen P. Ellner² and Jonathan M. Levine³

Abstract

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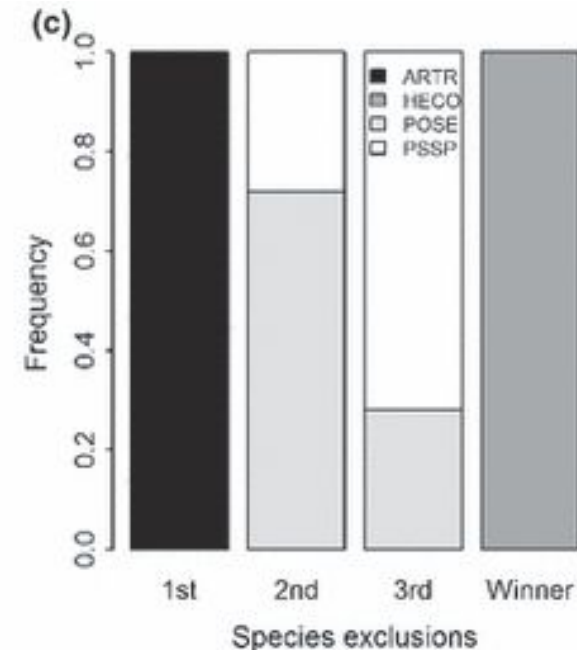
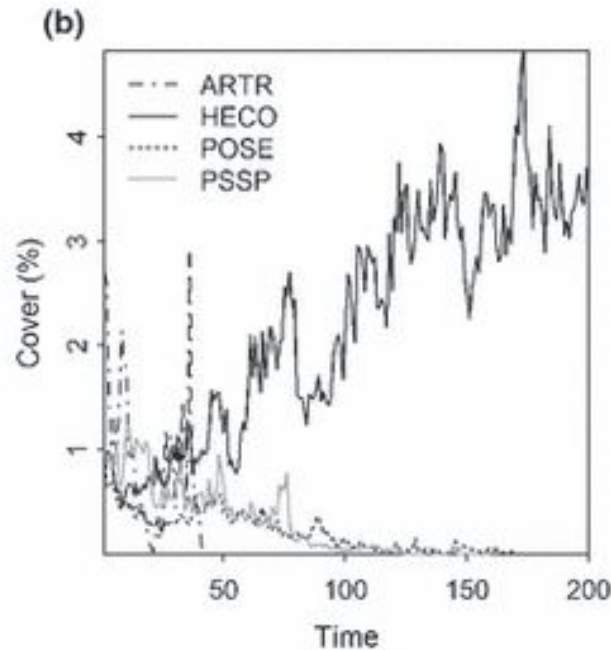
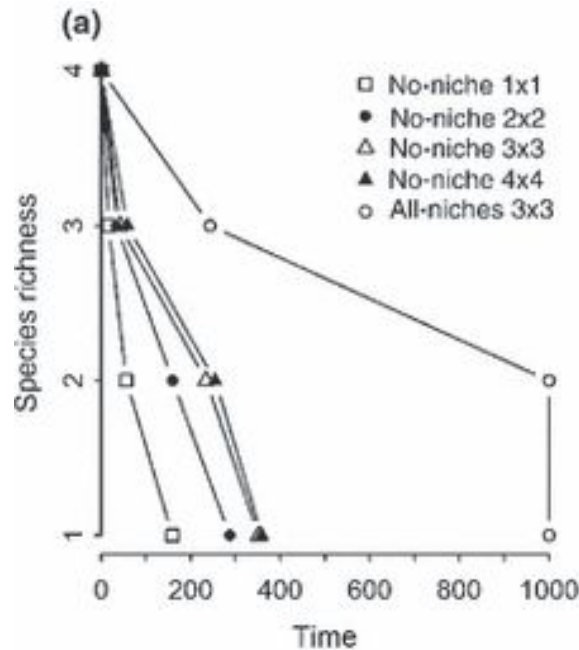
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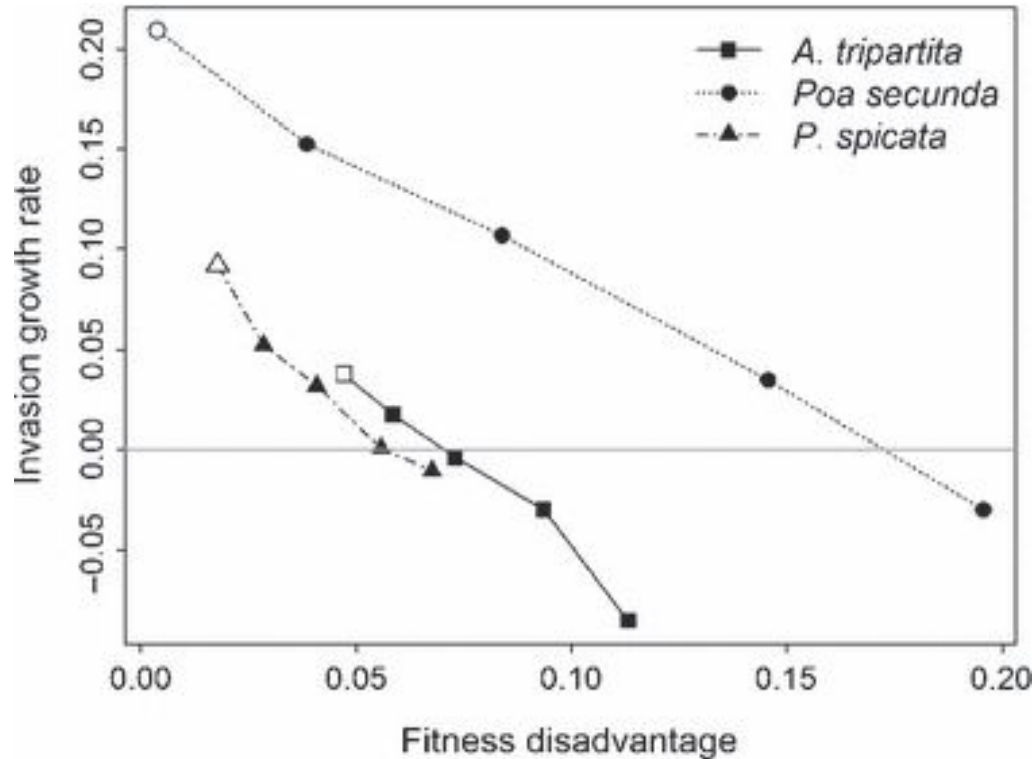
Coexistence of perennial plants: an embarrassment of niches

Abstract

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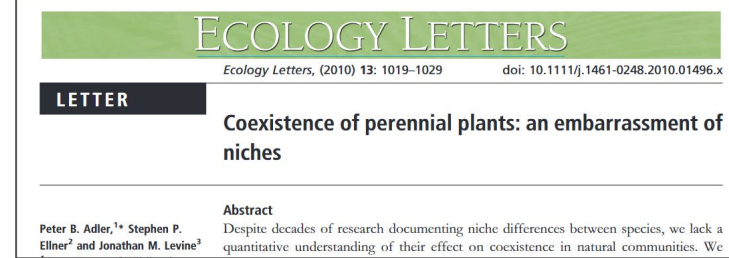
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Neutral theory and coexistence



Neutral theory and coexistence

- Strong support for niches
- Relatively strong
- May even underestimate niche strength



Neutral Theory

- Valuable as a null model
- Gets some things right, some things wrong
- Important in driving theory forward

Next class: Variable environments